

REVIEW PROBLEMS

Please note that this is NOT a practice exam – the length and style of questions aren't necessarily indicative of the length and style of the actual exam. This is just a review of problems addressing *some* of the concepts you learned so far.

QUESTION 1

Suppose that the observational unit in a study is a patient arriving at an emergency room in a given day. For each of the following, indicate whether it can legitimately be considered a variable or not. If it is a variable, classify it as categorical (and if it is binary) or quantitative. If it is not a variable, explain why not.

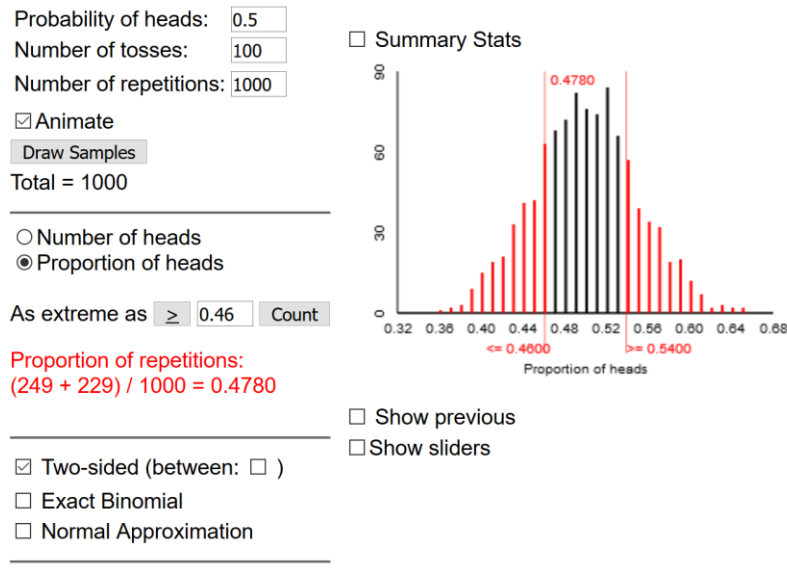
- a. Blood type
- b. Waiting time
- c. Mode of arrival (ambulance, personal car, on foot, other)
- d. Whether or not men have to wait longer than women
- e. Number of patients who arrive before noon
- f. Whether or not the patient is insured
- g. Number of stitches required
- h. Whether or not stitches are required
- i. Number of patients who are insured
- j. Assigned room number

QUESTION 2

Tennis players often spin a tennis racquet and observe whether it lands with the logo facing up or down, to determine who serves first. But is this really a 50-50 process, equally likely to land with the logo facing up or down? To investigate this, a tennis player spun his racquet 100 times, and he obtained 46 “up” and 54 “down” results. Does this provide much evidence against believing that spinning the racquet is really a fair 50-50 process?

- a. Produce a graph of these sample results.
- b. Identify the statistic and use an appropriate symbol to represent it.
- c. Identify the parameter in words using the context of the study.

- d. State the null and alternative hypotheses (both in words and symbols) corresponding to this research question.
- e. Describe how you could use a coin to conduct a simulation analysis of this study and its result. Give sufficient detail that someone else could implement this simulation analysis based on your description.
- f. Dorothy used the One Proportion applet to conduct a simulation analysis with 1,000 repetitions. The screenshot of her analysis is shown below. Was the analysis conducted correctly?



- g. Assume that the p-value is 0.48. Summarize the conclusion that you draw from this study based on your analysis. Make sure you explain the reasoning process behind your conclusion by referring to the (1) observed statistic, (2) p-value (3) justify strength of evidence against the null model (very strong, strong, moderate, weak) based on the p-value, (4) reject/fail to reject the null hypothesis, and (5) overall conclusion in context. Make sure you refer to the parameter in your answer.

QUESTION 3

One of the first times the U.S. Supreme Court considered statistical significance in an employment discrimination case was in *Hazelwood School District vs. United States (1977)*. The U.S. government sued the City of Hazelwood, a suburb of St. Louis, MO, on the grounds that it discriminated against African Americans in its hiring of school teachers. The evidence introduced noted that of the 405 teachers hired in 1972 and 1973 (the years following the passage of the Civil Rights Act), only 15 had been African-American. By comparison, according to 1970 census figures, of the almost 20,000 elementary and secondary teachers employed in the St. Louis area, 15.4% were African American. We want to decide whether the data on these 405 teachers is convincing evidence that the Hazelwood hiring process had a probability of a new hire being African American that was less than 0.154.

- a) What is one observational unit?

- b) What is the variable being measured in this study? What are the potential levels of this variable?
- c) If we define “success” to be hiring an African-American, calculate the statistic.
- d) Write a sentence describing the parameter of interest in the context of this study.
- e) Which symbol do we use to represent the parameter?
- f) State null and alternative hypothesis in symbols and numbers.
- g) In setting up a simulation in the One Proportion applet (see screenshot below), what are you going to use for the values of “probability of heads”, “number of tosses”, and “the number of repetitions”?

Describe process:

Probability of heads:
 Number of tosses:
 Number of repetitions:

☒ Show animation

Total Repetitions = 0

☐ Summary Statistics

Choose statistic:

☒ Number of heads
☐ Proportion of heads

Count samples

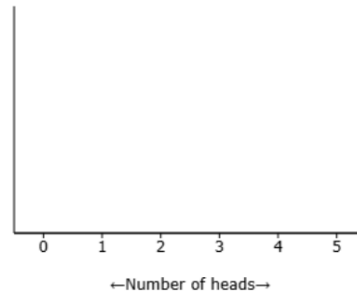
As extreme as

☐ Show previous results

☐ Show sliders

Options:

☐ Two-sided
☐ Exact Binomial
☐ Normal Approximation



- h) Suppose you conducted the simulation with a large number of repetitions and for each repetition you plotted the proportion of successes. What would be the center of the resulting dotplot (i.e. the simulated null distribution)?
- i) Suppose a student ran the simulation and got to the plot below. What number should you type into the red box to approximate the p-value? What sign should you use in the blue box ($\geq$ or $\leq$)?

Describe process:

Probability of heads: 0.154
Number of tosses: 405
Number of repetitions: 1000

☒ Show animation

Draw Samples

Total Repetitions = 1000

Choose statistic:

- ☐ Number of heads
☒ Proportion of heads

Count samples

As extreme as Count

Options:

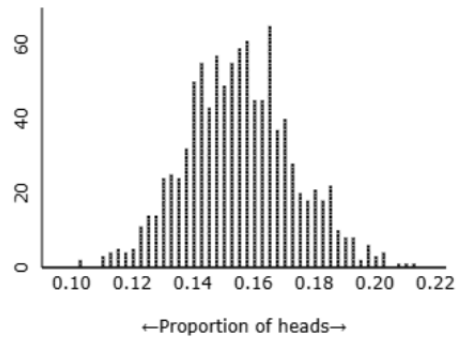
- ☐ Two-sided
☐ Exact Binomial
☐ Normal Approximation

Most recent results

Number of Heads = 69

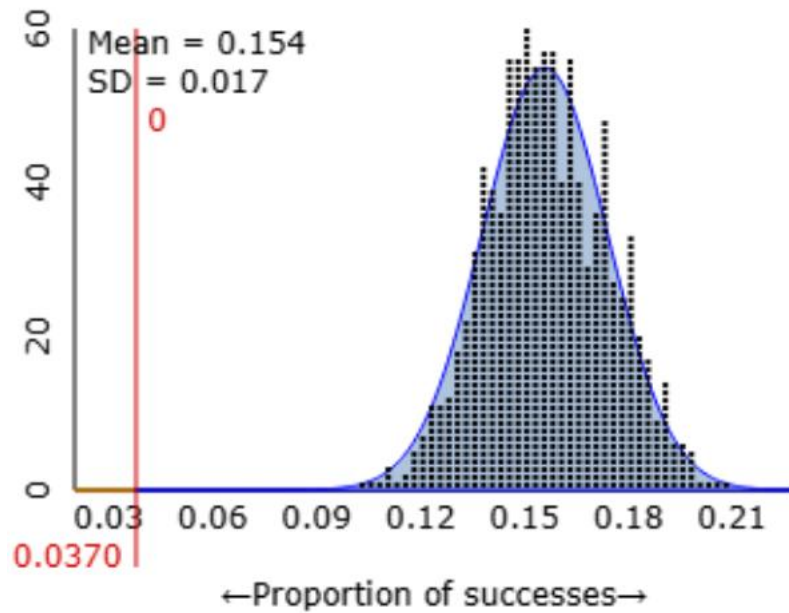
Number of Tails = 336

☐ Summary Statistics



- ☐ Show previous results
☐ Show sliders

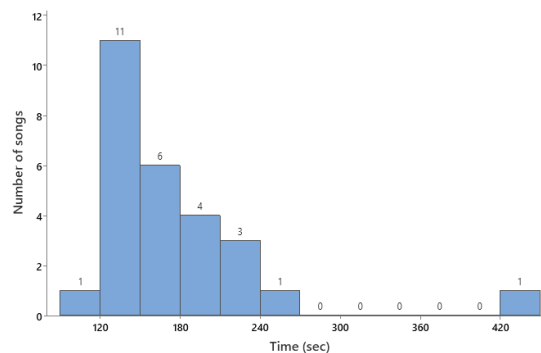
- j) Based on the screenshot above, which of the following would be the approximation of the p-value?
- A. 0.00
B. 0.50
C. 1.00
- k) Based on your answer to the previous question, what is the strength of evidence against the null hypothesis?
- l) Write a sentence interpreting the p-value in the context of this study.
- m) Are the validity conditions met to conduct a theory-based approach? Explain.
- n) Calculate the standardized statistic.
- o) Interpret the value you calculated in the previous question.
- p) The screenshot below shows the null distribution with the normal approximation overlaid. Does the normal model appear to reasonably predict the behavior of the simulated null distribution? Why do you think this is the case?



- q) Do the simulation-based p-value (3j), the normal approximation screenshot in the previous question (3p), and the standardized statistic from question (3n) agree with each other? Explain how you are deciding.
- r) The St. Louis City School District had recently followed a policy attempting to maintain a 50% African-American teaching staff. If you exclude the St. Louis City School District, then the proportion of eligible teachers in the St. Louis area that were African-American was 0.057 rather than 0.154. Find the new standardized statistic showing your work. Would you say there was stronger or weaker evidence against this null hypothesis. Explain your reasoning. You can assume that SD_{NULL} value is the same as before.

QUESTION 4

The following graph displays the lengths (in seconds) for the 27 Beatles songs that made it to a #1 ranking in the United States:



- a) What shape does this distribution have? (Circle your answer.)
a) Skewed to left

- b) Skewed to right
- c) Symmetric
- d) Bimodal

b) Which of the following is closest to the proportion of the 27 songs have a length between 2 and 3 minutes? (Circle your answer.)

- a) 0.2
- b) 0.4
- c) 0.5
- d) 0.6
- e) 0.8

c) Between what two (time) values does the median song length fall?

d) Between what two (time) values does the length of the longest song fall?

e) If the longest song is removed from the analysis, what would happen to the average (mean) of the song lengths? (Circle your answer.)

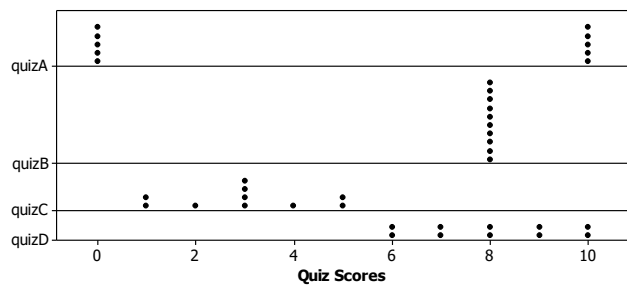
- a) Increase
- b) Stay the same
- c) Decrease

f) If the outlier were removed from the analysis, what would happen to the standard deviation of the song lengths? (Circle your answer.)

- a) Increase
- b) Stay the same
- c) Decrease

QUESTION 5

Consider the following four distributions of quiz scores:



Arrange these in order from smallest standard deviation to largest standard deviation. Just write in the letters A, B, C, D. Do not bother to calculate any standard deviations.

Smallest: Next-to-smallest: Next-to-largest: Largest:

QUESTION 6

Consider quiz scores for three new classes with the following distributions:

- Class E has thirty scores of 7.
- Class F has ten scores of 3, ten scores of 6, ten scores of 9.
- Class G has fifteen scores of 0 and fifteen scores of 10.

a) Which class has the largest standard deviation of quiz scores?

b) Which class has the smallest standard deviation of quiz scores?

QUESTION 7

In a study published in a 2007 issue of the journal *Preventive Medicine*, researchers found that people who used candy cigarettes as children were more likely to become smokers as adults, compared to people who did not use candy cigarettes as children. Identify one observational unit in this study. Circle your answer.

- A. A smoker
- B. A person who used candy cigarettes as a child
- C. A person
- D. A researcher

QUESTION 8

Researchers in Germany investigated a possible relationship between owning a pet bird and having lung cancer. They studied a sample of 239 lung cancer patients and also a sample of 429 people who did not have lung cancer, chosen to have similar characteristics to those with lung cancer. They asked all subjects whether they owned a pet bird in adulthood.

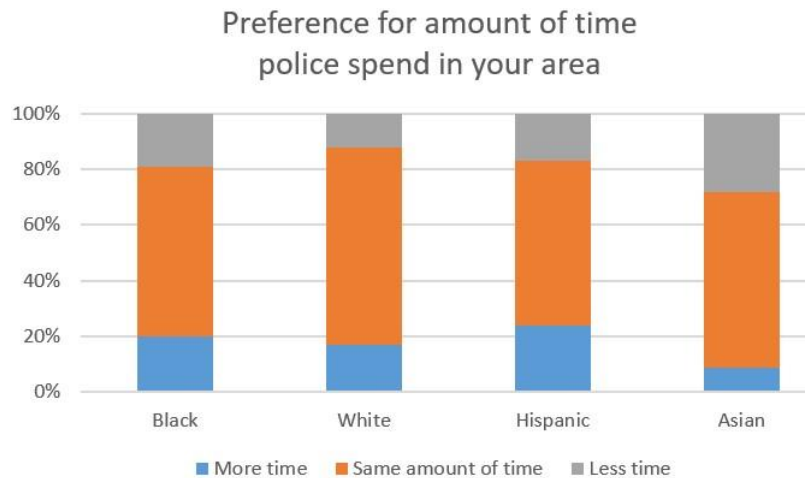
a) Identify the variables in this study.

b) What types of variables are these? Circle your answer.

- a) Both categorical
- b) Both quantitative
- c) One of each

QUESTION 9

The following graph was produced from data collected in a Gallup poll in June and July of 2020. Adult Americans were asked about their race and whether they would prefer that police spend more or less time in their area.



- How many variables are represented in this graph?
- What types of variables are represented in this graph?
- Which group has the largest percentage of people who prefer the police to spend less time in their area?
- Which group has the largest percentage of people who prefer the police to spend more time in their area?

QUESTION 10

Consider taking a sample of 10 words from the population of all words on a particular page of a popular book. You decide to select a sample by closing your eyes and pointing at the page 10 times. Would this constitute an unbiased sampling method?

- No, this sampling method would be biased toward large words
- No, this sampling method would be biased toward small words
- No, this sampling method would be biased toward nouns
- Yes

QUESTION 11

Suppose that a customer at the *In-n-Out* fast-food location in Arroyo Grande on September 30, 2020 constitutes an observational unit in a particular study. For each of the following, indicate whether it is a categorical variable, a quantitative variable, or not a variable.

- Whether or not the customer ordered a milkshake
 - Categorical variable
 - Quantitative variable
 - Not a variable
- Number of customers who ordered a milkshake
 - Categorical variable

- b) Quantitative variable
 - c) Not a variable
- c) Amount of the customer's bill
- a) Categorical variable
 - b) Quantitative variable
 - c) Not a variable
- d) Average amount of customers' bills
- a) Categorical variable
 - b) Quantitative variable
 - c) Not a variable
- e) Whether the customer placed the order at the drive-through or inside the restaurant
- a) Categorical variable
 - b) Quantitative variable
 - c) Not a variable

QUESTION 12

Suppose we took the set of all students in our class (STAT 1110) to be a sample. For which of the following variables would our sample likely be representative of the population of all Cal Poly students? (Circle your answer.)

- a) Major
- b) Sleep time last night
- c) Blood type